

Mani Amani

mamani@ucsd.edu | linkedin.com/in/maniamani | github.com/mamani828

Education

PhD in Electrical & Computer Engineering (Specialization in ML & Data Science), University of California, San Diego & San Diego State University • Advisors: Reza Akhavian and Nikolay Atanasov	Sep. 2024–Jun. 2029
MS in Computational Science (Transferred into PhD), San Diego State University • Advisors: Reza Akhavian	Sep. 2023–Jun. 2024
BS in Cognitive Science (Specialization in Machine Learning and Neural Computation), University of California, San Diego	Sep. 2018–Jun. 2023

Experience

Autonomy Engineer Intern, Michael Baker International – Pittsburgh, PA • Extended VidMap's structure-from-motion pipeline with multi-camera bundle adjustment, IMU-based gravity alignment, and support for fisheye and equirectangular cameras, increasing reconstruction acceptance rates from 40% to 85% on large-scale buildings, including schools and stadiums. • Developed an open-vocabulary asset extraction pipeline using OVI-MAP for semantic mapping of large indoor environments. • Implemented dynamic Gaussian splatting methods to model and account for moving objects in captured datasets.	Jun. 2026–Sep. 2026
AI Engineer Intern, AstreaX – Phoenix, AZ • Developed an end-to-end vision pipeline for detecting maritime hazards under low-light and nighttime conditions. • Built a synthetic data generation pipeline to improve dataset diversity, robustness, and model generalization. • Developed filtering methods for robust maritime hazard detection and tracking. • Trained a custom deformable transformer (DETR) model that achieved an 87.8% F1 score on held-out test benchmarks.	Jun. 2025–Sep. 2025
Research Assistant, San Diego State University & UC San Diego – San Diego, CA • Develop extensions of classical planning algorithms and safety formulations for robotic planning in high-dimensional latent spaces. • Developed LLM-guided robot planning and policy-modification methods using BIM, multi-heuristic A*, and semantic reasoning. • Designed a Q-learning framework for ergonomic optimization in human–robot bimanual handovers, published in <i>Automation in Construction</i> . • Developed LLM- and BIM-based methods for constructing semantic distance fields for robot navigation from natural-language user prompts. • Conducted numerical simulations and theoretical synchronization analysis of coupled nonlinear oscillators, published in the <i>International Journal of Bifurcation and Chaos</i> . • Implement bubble-cover planning algorithms with Control Barrier Functions (CBFs) on Clearpath Jackal robots for safe autonomous navigation. • Develop and maintain a ROS 2 Foxy autonomy stack for localization, mapping, and planning on Clearpath Jackal robots to support experimental evaluation of new planning algorithms. • Develop task-graph learning methods from scene graphs for semantic robot planning. • Lead and mentor a team of 10 undergraduate researchers developing robot-autonomy tutorials, contributing to the group's first annual robotic mapping workshop.	Jun. 2023–Present
Software Engineering Intern, San Diego Zoo Wildlife Alliance – San Diego, CA	Oct. 2024–Present

- Develop YOLO-based ensemble methods for detecting and tracking juvenile burrowing owls and potential predators.
- Design data pipelines for processing IMU measurements and analyzing elephant ethograms for behavioral monitoring.
- Develop infrared-vision algorithms for detecting and counting bats from robotic blimp imagery to support wildlife monitoring and disease-risk mitigation.

Publications

- **Amani, M.**, Beheshti, B., & Akhavian, R. (2026). “Bootstrapped LLM Semantics for Context-Aware Path Planning.” *ASCE International Conference on Computing in Civil Engineering (i3CE)*. arXiv:2511.11967
- **Amani, M.**, & Akhavian, R. (Accepted, 2026). “Digital Twin-Guided Robot Path Planning: A Beta-Bernoulli Fusion with Large Language Model as a Sensor.” *Journal of Computing in Civil Engineering*. arXiv:2509.20709
- Dai, Z., Fan, T., **Amani, M.**, Seo, J., Lee, K. M. B., Oh, H., & Atanasov, N. (2026). “Kernel-SDF: An Open-Source Library for Real-Time Signed Distance Function Estimation using Kernel Regression.” *IEEE Robotics and Automation Letters*. doi:10.1109/LRA.2026.3719228
- Akhavian, R., **Amani, M.**, Mootz, J., Beheshti, B., & Ashe, R. (2026). “Semantic-Gravity Iterative Closest Point and AI-assisted navigation for Robot-Ready Construction Digital Twins.” *Automation in Construction*, 190, 107134. doi:10.1016/j.autcon.2026.107134
- **Amani, M.**, & Akhavian, R. (2025). “Safe and trustworthy robot pathfinding with BIM, MHA*, and NLP.” *Construction Robotics*, 9, Article 18. doi:10.1007/s41693-025-00161-1
- Ashe, R., **Amani, M.**, Sommerfield, C., & Akhavian, R. (2025). “A Customized Quadruped Control and Sensor Integration Framework with a Construction Worker Activity Recognition Module.” *Computing in Civil Engineering* 2025, 708–712. doi:10.1061/9780784486443.077
- **Amani, M.**, & Akhavian, R. (2024). “Intelligent ergonomic optimization in bimanual worker-robot interaction: A Reinforcement Learning approach.” *Automation in Construction*, 168, 105741. doi:10.1016/j.autcon.2024.105741
- Palacios, A., In, V., & **Amani, M.** (2024). “Disorder-Induced Dynamics in Complex Networks.” *International Journal of Bifurcation and Chaos*, 34(5), 2430010. doi:10.1142/S0218127424300106
- **Amani, M.**, & Akhavian, R. (2024). “Adaptive Robot Perception in Construction Environments Using 4D BIM.” *Computing in Civil Engineering* 2024, 679–687. doi:10.1061/9780784486115.072

Projects

A Quadruped High-Level Control Module for Worker Activity Recognition Sep. 2024

- Developed a high-fidelity 3D environment for BIM-centric mission planning with a quadruped robot.
- Created a modular, open-source software package for streamlined quadruped mission planning in AEC.
- Designed a novel self-labeling mechanism for dynamic data generation in human activity recognition.
- Source code available on GitHub: github.com/mamani828/GO1_GPT_UNITY; manuscript under review at the International Conference on Computing in Civil Engineering.

Intelligent Ergonomic Optimization in Bimanual Worker-Robot Interaction Sep. 2024

- Designed a fast-converging Q-learning algorithm for Rapid Entire Body Assessment (REBA) scoring.
- Developed a VR framework using inverse kinematics tailored to individual worker physiques.
- Achieved a $2.5\times$ improvement in ergonomic metrics post-optimization; published in *Automation in Construction*.
- Source code available on GitHub: github.com/mamani828/RL-REBA.

MOLE Annotated May. 2024

- Created a tutorial for beginners on numerical PDE solvers, with a focus on mimetic methods.
- Demonstrated the numerical solution of the time-dependent Schrödinger equation using mimetic discretization techniques.

- Adopted by the MOLE developer team at SDSU as a resource for users.
- Available on GitHub and GitHub Pages: mamani828.github.io/Annotated_MOLE.

Technologies

Languages:

- **C++:** Real-time robotics (ROS, ROS2), high-performance computing (multithreading, OpenMP, MPI)
- **Python:** Machine learning and data analysis (NumPy, PyTorch, pandas), scripting and prototyping
- **C#:** Unity scripting for 3D simulations and robot visualization
- **CUDA-C & ROCm:** GPU acceleration for PDE solvers, SLAM, and parallel computing

Tools/Frameworks:

- **ROS/ROS2:** Developing modular robotics pipelines for mapping, localization, motion planning
- **Unity:** Building custom 3D environments for robotics simulations and scenario testing
- **PyTorch:** Designing and training deep neural networks for perception and control
- **Gazebo/RViz:** Simulation and visualization of mobile robots for rapid prototyping
- **Docker & GitHub CI/CD:** Containerized deployments and continuous integration workflows

Honors and Awards

- Recipient of the SDSU University Graduate Fellowship (UGF) (2026-2028)
- Recipient of the Achievement Rewards for College Scientists (ARCS) Scholarship (2026-2027)
- Recipient of the NSF-Funded S-STEM ASSICS Scholarship, SDSU (2023–2025)
- Recipient of the SDZWA & San Diego Foundation Fellowship (2024)
- Runner-up Poster Award, SDSU Graduate Research Symposium (2025) *“Intelligent ergonomic optimization in bimanual worker-robot interaction: A Reinforcement Learning approach”*
- Best Poster Award, SDSU Graduate Research Symposium (2024) *“BIM-based Safe and Trustworthy Robot Pathfinding using Scalable MHA* Algorithms and NLP”*
- Provost Honors, UCSD (2018, 2020)